

Texas College of Management & IT

Bachelor of Computer Science (Hons) (Network Technology & Cybersecurity)

Assignment Title: " The Digital Identity Project "

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Course: Web Desgining (WBD - 113)

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STUDENT DECLARATION

I declare that this mini project is my original work and completed by me.

Student Signature: Animesh Babu Karki

Date: 27/07/2026

FACULTY EVALUATION

Marks Obtained: _____

Comments:

Faculty Name: _____

Signature: _____

Date: _____

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Introduction

This report walks through how I planned, designed, and built my personal portfolio website for the Digital Identity Project. The goal was straightforward: make a site that shows off who I am, what I can do, and what I've built so far, while proving I actually understand the web dev stuff we covered in class, like semantic HTML, CSS layout tools like Flexbox and Grid, typography, colour theory, responsive design, and a bit of JavaScript for interactivity.

The site is aimed at employers, collaborators, and recruiters in product design, networking, and cybersecurity. So every choice I made, like section order, colours, and fonts, was about looking professional and trustworthy while hinting at my unique qualities, as fast as possible. Most people decide whether to stick around a webpage in a few seconds, so I wanted those first seconds to count.

Project Objectives

Before I started coding, I jotted down a short list of goals, so I knew what I was actually trying to do. These came straight from the assignment brief, and I kept coming back to them to check I hadn't drifted off track:

- Build a responsive, single-page-style personal portfolio that adapts fluidly across desktop, tablet, and mobile viewports.
- Use CSS Flexbox and Grid for a clean, consistent layout.
- Apply CSS Flexbox and Grid systems to construct a clean, consistent, and adaptable layout.
- Implement basic JavaScript interaction to enhance usability without compromising performance.
- End up with a professional, recruiter-ready site that shows off skills, experience, and projects, with clear contact info.

Core Understanding and Planning

Before writing any code, I spent time planning the site structure, like a real developer would before starting a project. This stage turned the vague idea of "building a portfolio" into concrete decisions about layout, colour, and type.

The site follows a top-to-bottom story in this order:

Header/Navigation → Hero → About → Activity → Highlighted Work → Side Projects → Ideas → Contact/Footer.

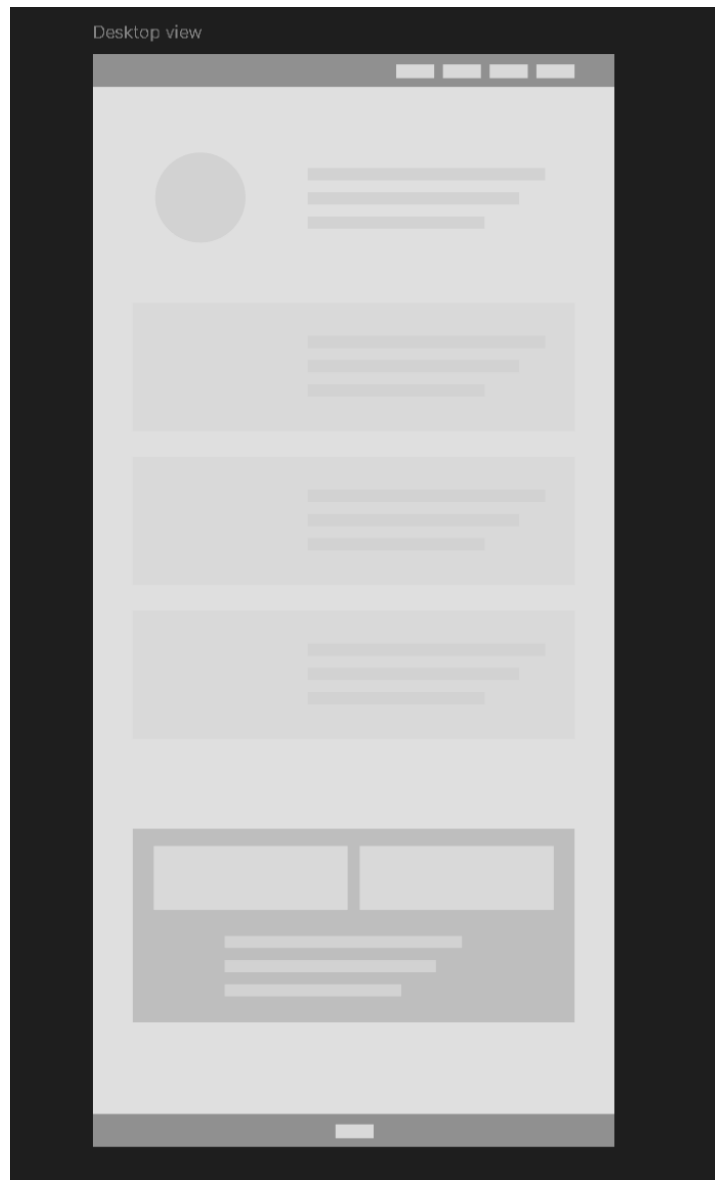


Figure 1. Desktop Wireframe

I sketched this out as a simple ‘Wireframe’ first, shown above, before touching any code. A wireframe is just a rough, low-detail sketch of where everything will go on the page. It let me test the ordering of the sections and move things around cheaply, without having to rewrite CSS every time I changed my mind.

I chose this order deliberately to work with the "F-Layout" scanning pattern. Research shows most web readers, including recruiters skimming a candidate's page, scan in two horizontal movements across the top, then drop into a vertical scan down the left side.

To take advantage of that, I put the most important credibility signals along that top band and left edge: my name and title sit in the upper-left of the header, navigation runs

horizontally across the top, and the hero section underneath introduces my photo, role, focus areas, tech stack, experience, and location in a left-aligned list. So even a recruiter who only reads the top and skims down the left margin will still catch my name, discipline, and key qualifications before deciding whether to dig deeper.

Colour Palette

The brief recommended at most three colours (primary, secondary, and accent) and suggested dark mode as one option. I went with a light, high-contrast palette instead. A bright background with one saturated accent colour reads as clean and approachable for a design and security audience, while still sticking to the three-colour limit:

- Primary (background): White, #FFFFFF
- Secondary (text and structure): Dark grey/near-black body text, with supporting neutral greys of  #F4F4F4,  #EEEEEE,  and #999999 used for section backgrounds, borders, and muted text.
- Accent: Blue,  #2F6FED, used sparingly for links, highlighted keywords, and interactive states so that it draws the eye without overwhelming the page.

Keeping it this simple made the page feel calm instead of chaotic, and the blue accent actually stood out whenever it appeared because it wasn't competing with five other bright colours. If I'd used more, the important clickable stuff could've easily gotten lost in the noise.

Typography

For headings, I chose Tomorrow (Google Fonts), a clean, modern sans-serif that gives the page a confident feel without shouting. For body text, I used Geist Mono (Google Fonts), a monospace font where every letter takes up the same width, like an old typewriter. I liked that it hints at the technical and cybersecurity side of what I do while staying easy to scan.

Both fonts are self-hosted using CSS `@font-face` instead of loading from a third-party CDN. That removes an external network dependency and gives a small performance and privacy boost.

Tomorrow

Designed by [Tony de Marco](#), [Monica Rizzoli](#)

Sans Serif — Geometric

Feeling — Futuristic

Feeling — Stiff

Feeling — Rugged

Appearance — Techno

Feeling — Business

Feeling — Vintage

Feeling — Sincere

Feeling — Loud

Whereas disregard and
contempt for human rights have
resulted

Figure 2: Font 'Tomorrow'

Geist Mono

Designed by [Andrés Briganti](#), [Mateo Zaragoza](#), [Guillermo Rauch](#), [Evil Rabbit](#), [José Rago](#), [Facundo Santana](#) | [@vercel.com/font](#)

[Learn more about variable fonts](#) 

Appearance — Monospaced

Technology — Variable

Feeling — Rugged

Feeling — Calm

Feeling — Futuristic

Feeling — Business

Feeling — Vintage

Feeling — Loud

Whereas disregard and
contempt for human rights
have resulted

Figure 3: Font 'Geist Mono'

Development Steps and Code Structure

With the plan in place, development proceeded through four broad stages: semantic markup, layout and styling, interactivity, and responsiveness.

Semantic HTML Structure

Instead of wrapping everything in generic div (which don't tell a browser or screen reader anything about what's inside), I used semantic HTML5 tags wherever they made sense, like header, nav, main, section, and footer. Each major section, including hero, About, Activity, Highlighted Work, Side Projects, and Ideas, got its own section tag with a proper heading. The nav links directly to each section using anchor links like #about, #career, #work, #projects, and #ideas. This makes the page easier for screen reader users and helps search engines understand what each part of the page is about.

```
28 <section id="about">
29   <div class="about-flex">
37     <ul>
38       <li><strong>Focus:</strong> <span class="fw-medium">Product Design</span> and <span class="fw-medium">Cyber
39       <li><strong>Stack:</strong> Adobe Creative Suite (Ps, Ai, Ae, etc.), Figma, Python, and more.</li>
40       <li><strong>Experience:</strong> 2+ years in Graphic Design, currently learning Networking & Cybersecurity
41       <li><strong>Location:</strong> <span class="fw-medium">Kathmandu, Nepal</span></li>
42     </ul>
43     <p>
44       <a href="https://github.com/ekka-np" target="_blank">GitHub</a> |
45       <a href="https://www.instagram.com/animeshkarki_" target="_blank">Instagram</a> |
46       <a href="mailto:animeshbabukarki@gmail.com">Email Me</a>
47     </p>
48   </div>
49 </div>
50 </section>
51
52 <!-- Activity Section -->
53 <section id="career">
54   <h2>Activity</h2>
55   <p class="section-desc">Overview of my current academic and professional duties.</p>
56   <ul class="career-list">
57     <li>
58       <span class="meta meta-date">November 2024 - Present</span>
59       <span class="card-title">BCS</span>
60       <span class="meta meta-org">Texas College of Management & IT</span>
61       <p>currently pursuing a Bachelor's in Computer Science, specializing in Networking and Cybersecurity.</p>
62       <p>I'm driven by the challenge of staying ahead of the ever-evolving cyber threats of tomorrow. My primary a
63       <span class="meta meta-skills">Secure Architecture, Secops, Threat Hunting</span>
64     </li>
65     <li>
66       <span class="meta meta-date">October 2025 - Present</span>
```

Figure 4. HTML code

CSS Layout and Styling

I used CSS Flexbox for one-dimensional groupings like the nav bar, social link row, and certification list. CSS Grid handled two-dimensional layouts like the hero section and project galleries, where an image column and text column need to align across a row. Throughout the site, I kept spacing consistent, stuck to my small colour palette, and used only

the one accent colour. The result is a clear visual hierarchy. It's obvious what to look at first without needing heavy borders, boxes, or drop shadows everywhere.

JavaScript Functionality

I used vanilla JavaScript sparingly, only where it actually improved usability. A back-to-top button appears after you scroll past the hero section and smoothly scrolls you back up, which helps on a long single-page layout. The second feature is a simple overlay on project screenshots that makes it harder to right-click and save the original images. Some of that design and photography work is mine, and I'd rather people not casually copy and save it.

Responsive Design Implementation

CSS media queries reflow grid sections into a single column on narrow screens, stack the nav for touch input, and resize typography and images so content stays readable without horizontal scrolling. I checked the layout at common breakpoints for phone, tablet, and desktop widths.

Testing and Quality Assurance

Once the site was built, I went through and tested it manually in both Chrome and Firefox to make sure it behaved the same way in each browser. I checked that every internal navigation link scrolled to the right section, that the page resized properly across different screen widths, that the external links to my GitHub, Instagram, and the pages of the organisations I work with all pointed to the right place, that the mailto link correctly opened an email addressed to me, and that every image and video on the page actually loaded instead of showing a broken link.

Challenges and Solutions

Not everything worked on the first try. The three biggest headaches were: getting project screenshots to size correctly when they all had different aspect ratios, keeping the footer lined up across different screen sizes, and getting the spacing between sections to feel right. Early versions felt either too cramped or way too empty. None of these were one-and-done fixes. I had to go back several times and tweak things like CSS Grid track sizes, the object-fit property on images (which controls how an image is cropped or scaled inside its

box), and padding above and below each section before the layout finally felt stable and consistent..

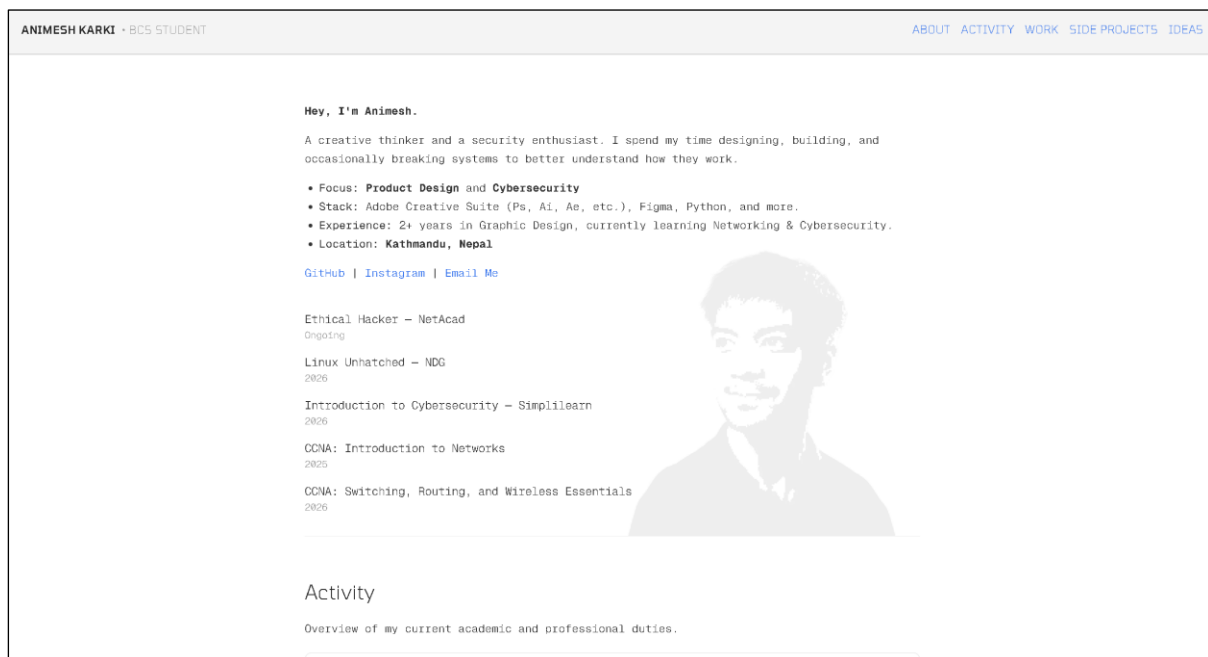


Figure 5. Final Website's Hero Section

Future Improvements

The site is finished for this assignment, but there are a few things I would like to add or improve if I keep working on it afterwards:

- Adding a dedicated dark-mode toggle so visitors can switch between the current light theme and a dark aesthetic.

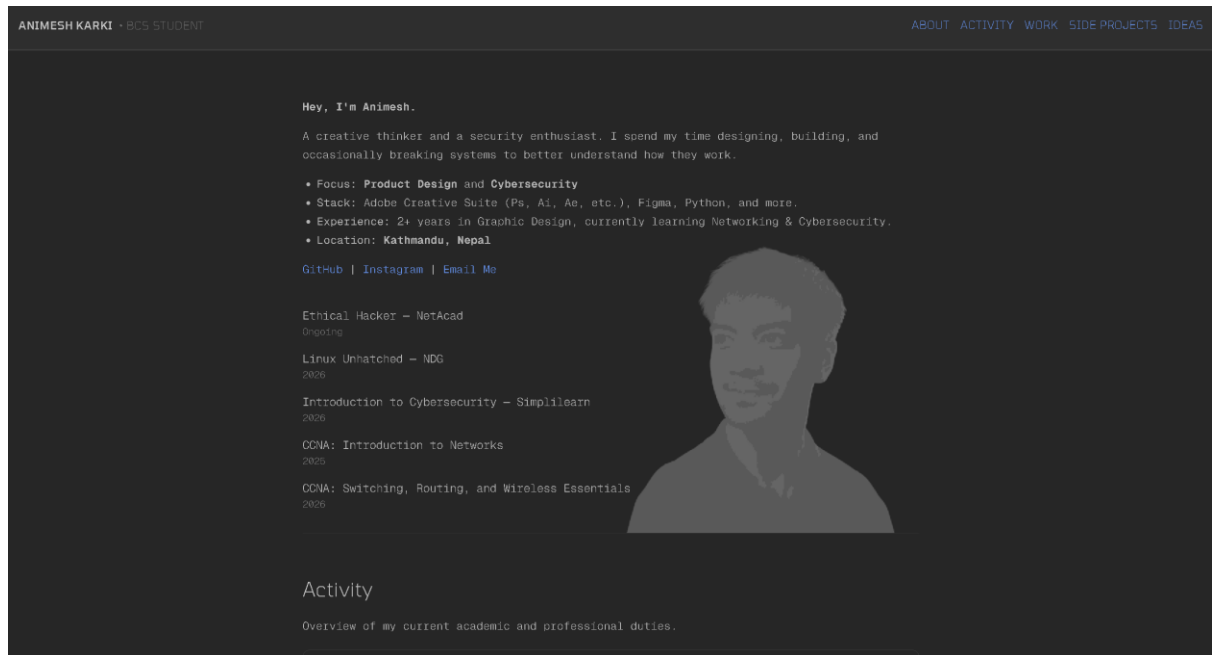


Figure 6. Idea for a 'Dark' mode for the Webpage

- Introducing a hamburger-style navigation menu for small screens to further simplify the mobile header.
- Improving accessibility through additional ARIA labels and a full keyboard-navigation audit.
- Adding subtle, performance-conscious animations to guide attention during scrolling.
- Doing more performance work, such as compressing images further and lazy loading, so my Lighthouse performance (Currently 83 on Mobile, See Appendix) score gets even closer to a perfect 100.

Submission Requirements

The following materials are provided in support of this submission, in accordance with the assignment's grading criteria.

GitHub Repository

The project's version-controlled source code was initialised as a Git repository and pushed to GitHub for public review at the following address:

<https://github.com/ekka-np/web-dev>

Live Deployment

The site is deployed and publicly accessible via GitHub Pages at the following address:

<https://ekka-np.github.io/web-dev/>

Screenshots

Desktop View

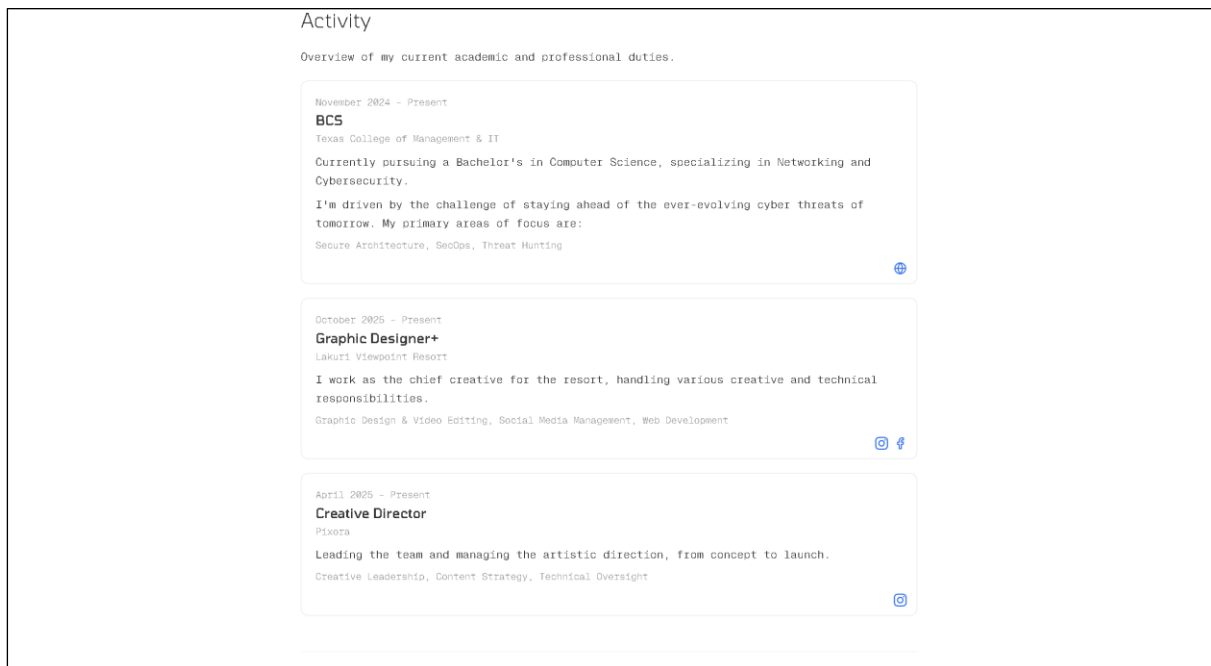


Figure 7. Desktop View (1)

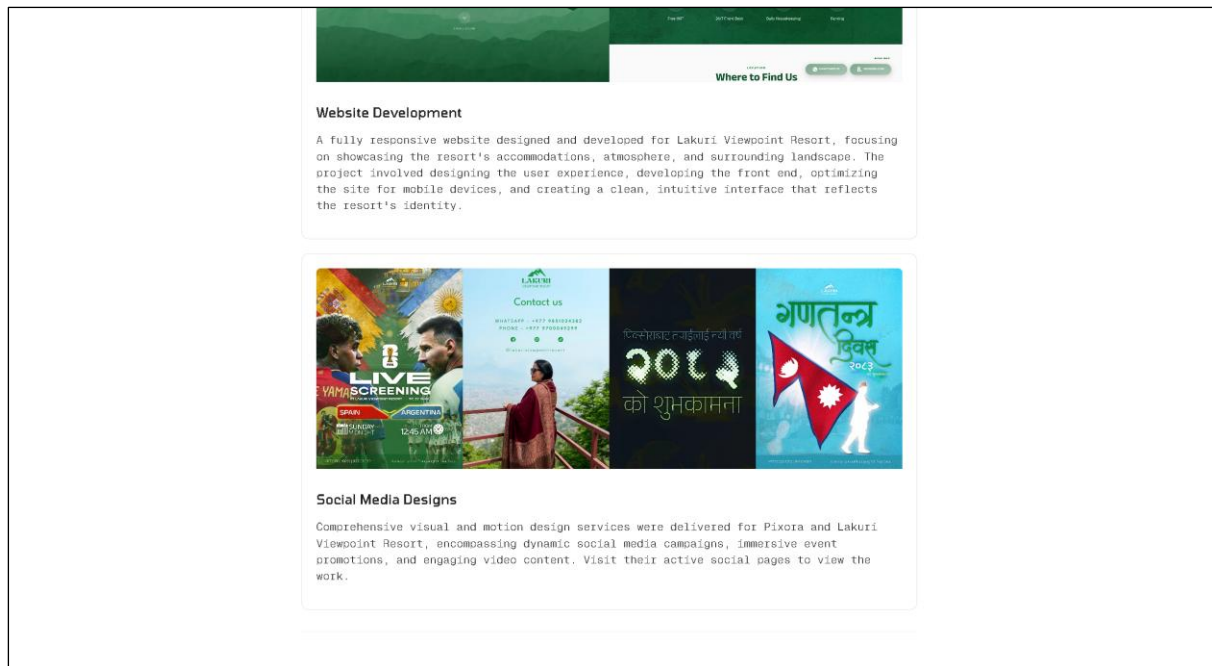


Figure 8. Desktop View (2)

Mobile View

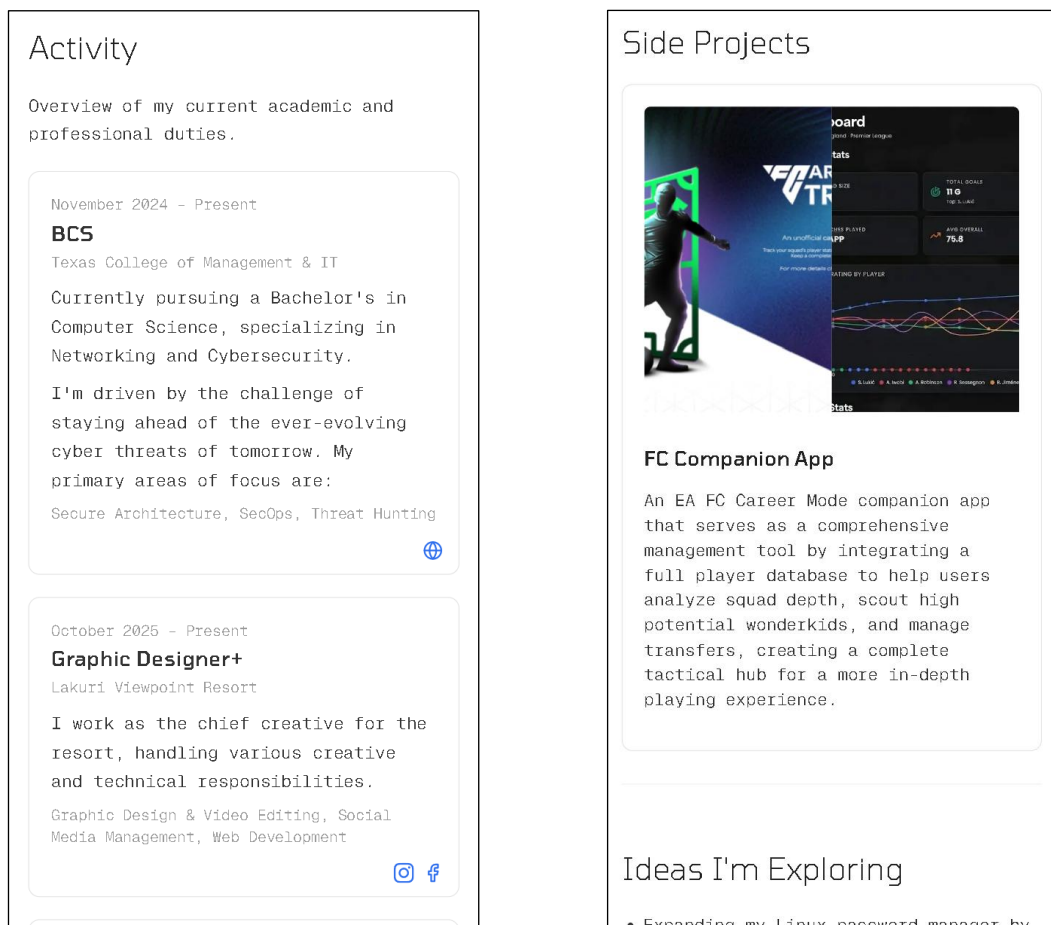


Figure 9. Mobile Views

Conclusion

Working on the Digital Identity Project meant taking everything I learned in class, semantic HTML, layout systems, typography, colour theory, and responsive design, and actually turning it into something real: a portfolio website that a recruiter or a collaborator could genuinely use to learn about my skills and my projects. Planning the page around the F-Layout pattern before I wrote any code meant that the most important parts of my story would land exactly where a visitor's eyes naturally go, and keeping my colours and fonts simple made the whole site feel professional and easy to follow rather than cluttered. Between the finished website, the GitHub repository behind it, and the live deployment, I feel like this project meets everything the assignment brief asked for.

Appendix

Website's Lighthouse Score

The completed portfolio website was evaluated using Google Lighthouse to assess its overall quality in terms of performance, accessibility, best practices, and search engine optimisation. The evaluation on the Mobile view produced an overall *Performance score of 100, Accessibility score of 93, Best Practices score of 100, and SEO score of 100*, demonstrating that the website is highly optimised, adheres to modern web development standards, and provides an efficient and reliable user experience.

The performance metrics further reinforce these results, with a *First Contentful Paint (FCP) of 0.3 seconds, Largest Contentful Paint (LCP) of 0.4 seconds, Speed Index of 0.5 seconds, Total Blocking Time (TBT) of 0 ms, and a Cumulative Layout Shift (CLS) of 0.019*. These values indicate that the website loads rapidly, remains responsive throughout user interaction, and maintains excellent visual stability.

Although the website achieved outstanding overall results, the Lighthouse report identified a small number of opportunities for further enhancement, including improving image delivery efficiency, reducing unused JavaScript, and refining browser caching to optimise resource loading. Overall, the Lighthouse evaluation confirms that the website delivers excellent performance while complying with recognised accessibility, performance, SEO, and web development best practices.

The original HTML and PDF files are also in the github repository.

<https://github.com/ekka-np/web-dev/tree/main> and

<https://github.com/ekka-np/web-dev/tree/docs>

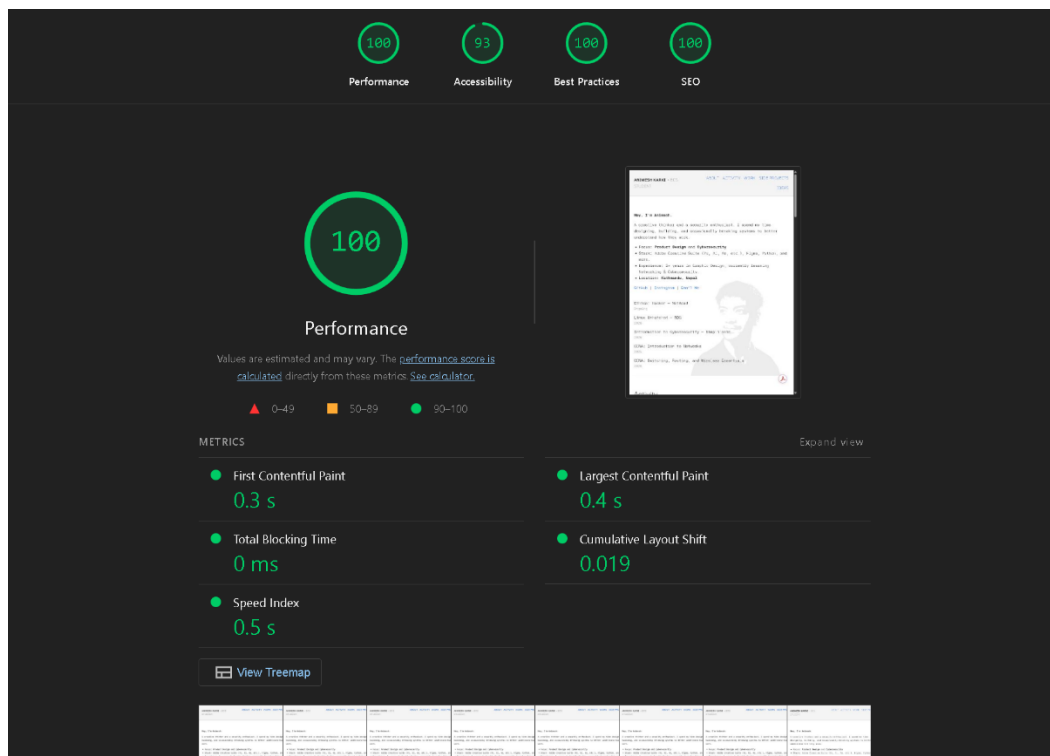


Figure 10. Webpage's Browser Lighthouse Score

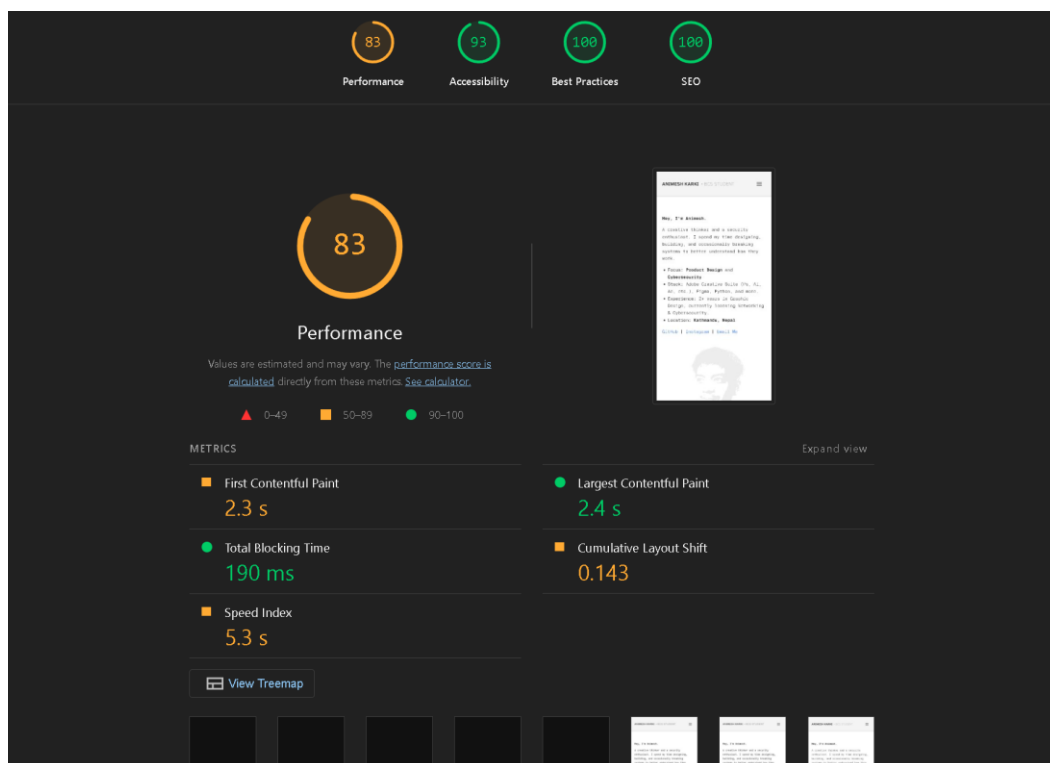


Figure 11. Webpage's Mobile Lighthouse Score